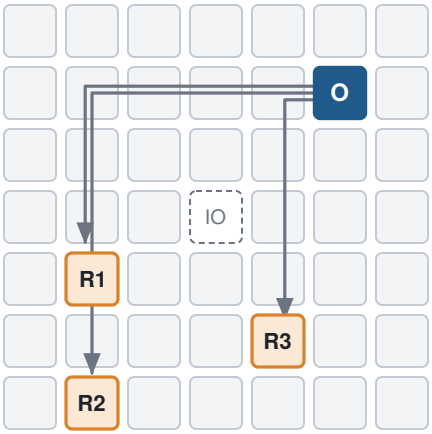
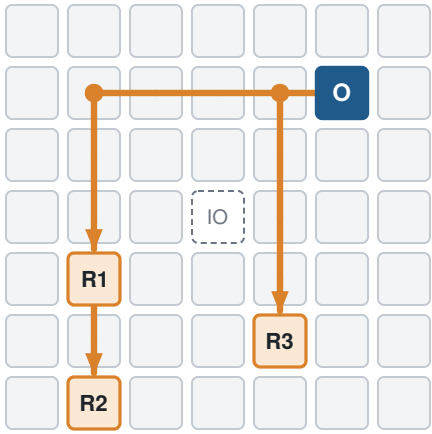
(a) Baseline



owner reads the line 3 times,
sends 3 separate responses

(b) Two-sided merge



owner reads once; one response forks along the XY tree (dots)

(c) Table entries

Requester in-flight table (IFT), per GPM

fingerprint	dst GPM	waiter list ptr	V

+ full line tag kept in a side array: checked before any merge

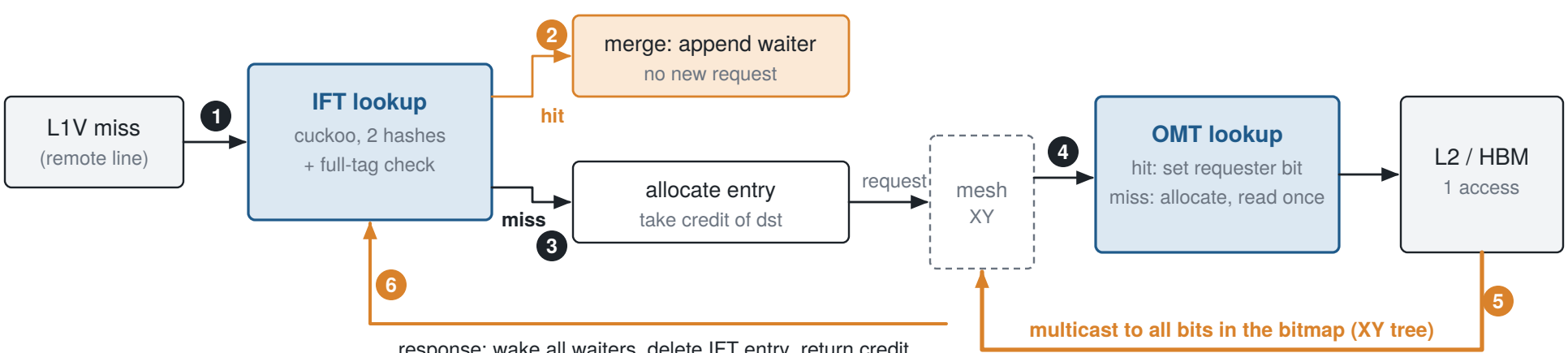
Owner merge table (OMT), per GPM

fingerprint	requester bitmap (48 b)	state

Distance-aware credits (window W split by RTT)

1 hop	RTT 155 ns
3 hops	RTT 298 ns
6 hops	RTT 514 ns
12 hops	RTT 945 ns

credits proportional to RTT: every destination can reach
its bandwidth (Little's law)

(d) Requester side (RDMA of each GPM)

(e) Owner side